**DELHI PUBLIC SCHOOL NEWTOWN**

**SESSION: 2023-2024**

**HALF YEARLY EXAMINATION**

**CLASS: X** — **FULL MARKS: 100**

**SUBJECT: COMPUTER APPLICATIONS [SET A]** — **TIME: 2 HOURS**

***Answers to this Paper must be written on the paper provided separately.***

***You will not be allowed to write during the first 15 minutes.***

***This time is to be spent in reading the question paper.***

***The time given at the head of this Paper is the time allowed for writing the answers.***

***This paper consist of nine printed pages. This Paper is divided into two Sections.***

***Attempt all questions from Section A and any four questions from Section B.***

***The intended marks for questions or parts of questions are given in brackets[].***

## SECTION A(40 Marks)

*(Attempt all questions from this Section)*

**Question 1**

**Choose the correct answer and write the correct option.**

**(i) The concept of Abstraction in Java is:** **[1]**

(a) Binding data and functions together
(b) Hiding inner complexity and providing usable interfaces
(c) Reusing of the code
(d) Making methods constant

**(ii) Point out the valid object name** **[1]**

(a) oop1 (b) 1oop (c) Oop 1 (d) Oop#1

**(iii) The method to check if a character is an alphabet or not is:** **[1]**

(a) isLetter(char) (b) isAlpha(char)
(c) isUppercase(char) (d) isLowercase(char)

**(iv) Predict the output: System.out.println(Math.ceil(-1.5+ Math.abs(-2.7)));** **[1]**

(a) -4.0 (b) -4 (c) -3 (d) -3.0

**(v) What is the output when the following method has been executed?** **[1]**

```
import java.util.*;
public class vedant
{  int i=0, s = -2;
   void funcA()
   {  for (int k = 0; k < 3; k++)
      {  do
         { - -i;  ++s; } while (i > k); }
         System.out.println(i);   }   }
```

**(a) -2 (b) -1 (c) 0 (d) -3**

**(vi) Evaluate the expressions: S = ++m/5 !=5 || n*5 >= n*n++, where, m=-2 and n= -1** [1]

**(a) true (b) false (c) error (d) no output**

**(vii) How many times the inner loop is executed?** **[1]**

```
for(int k=1;k<=2;k++)
{ for(int m=1;m<=k;m++)
  { System.out.println(m);}}
```

**(a) 4 times (b ) 3 times (c) 2 times (d) 5 times**

**(viii) Which of the following is not applicable for a constructor function?** **[1]**

**(a) It has the same name as the class.**

**(b) It has no return-type**

**(c) It is automatically called when object is created.**

**(d) It can be invoked using an object like any other member function.**

**(ix) Predict the output of the given code:** **[1]**

```
String n1 = "46", n2 = "64";
int total = Integer.parseInt(n1) + Integer.parseInt(n2);
System.out.println("The sum of " + "46 " + "and" + " 64" + " is " + total);
```

**(a) The sum of 46 and 64 is 110**

**(b) The sum of 46 and 64 is 4664**

**(c) No output**

**(d) Compilation error**

**(x) Generally, a computer language uses either compiler or interpreter to convert source code into object code. But Java, a case sensitive language, uses both the translators (i.e., compiler as well as interpreter). Firstly, compiler converts the Java source code into an intermediate code which is further converted into machine code with the help of an interpreter. Based on the above discussion, answer the following questions:**

**(a) Which of the following statement is true? [1]**

1. **Java compiler is a hardware.**
2. **Java compiler is a software but interpreter is a hardware.**
3. **Both the translators are hardware.**
4. **Both the translators are software.**

**(b) The intermediate code is referred to as: [1]**

**1. Byte code 2. Assembly code 3. ASCII code 4. Unicode**

**(c) Java interpreter is also known as: [1]**

**1. Artificial machine 2. Virtual machine 3. Real machine 4. Normal machine**

**(d) What is meant by a case sensitive language? [1]**

1. **It understands only lowercase letters.**
2. **It understands only uppercase letters.**
3. **It distinguishes between lowercase and uppercase letters.**
4. **It doesn't distinguish between lowercase and uppercase letters.**

**(xi) What will be the output of the Java Program with many constructors? [1]**

```
public class Constructor
{  Constructor(boolean a)
    { System.out.println("MODEM="+ a ); }
   Constructor(float a)
    { System.out.println("ROUTER=" + a);   }
 public static void main()
    {   Constructor con1 = new Constructor(50);
        Constructor con2 = new Constructor(false); }}
```

**(a) ROUTER=50.0 (b) ROUTER=50 (c ) Compiler error (d) Semantic error**

**MODEM=false MODEM=false**

**(xii) Name the type of error that occurs for the following statement: [1]**

**System.out.println(Math.sqrt(24 - 25));**

**(a) Syntax error (b) run time error (c) logical (d) no error**

**(xiii) Assertion(A): In java, objects interact with one another by passing messages.**

**Reason(R): This behavior of the object is depicted through methods or functions. [1]**

**(a) Both Assertion (A) and Reason (R) are true and (R) is a correct explanation of (A)**

**(b) Both Assertion (A) and Reason (R) are true and (R) is not a correct explanation of (A)**

**(c) Assertion (A) is true and Reason (R) is false**

**(d) Assertion (A) is false and Reason(R) is true**

**(xiv) Functions are the built-in modules that can be used in our program to perform a specific operation. There are some mathematical functions that are used for specific functions. For example, we can use Math.sqrt( ) function to return square root of a positive number, Math.random( ) and Math.ceil( ) to return random number in a range of numbers to return the next higher integer number. Based on the above discussion, answer the following:**

**(a) What will be displayed, if the following statement is executed? [1]**

**System.out.println(Math.round(-2.63));**

**1. -3 2. -2.0 3. -1 4. -2.0**

**(b) Math.random( ) function returns a random number in the range: [1]**

**1. between 0 and 1 2. between -1 to 0 3. between -1 to 1 4. None of the above**

**(c) Which of the following function will return the next higher integer number? [1]**

**1. Math.max( ) 2. Math.min( ) 3. Math.floor( ) 4. Math.ceil( )**

**(d) Which of the following statement is true for Math.max( )? [1]**

**1. It will always result in int type value.**

**2. It will always result in double type value.**

**3. The resultant data type will depend upon the type of arguments.**

**4. It will return float type, if arguments are double type.**

**Question 2**

**(i) How is autoboxing different from unboxing? Show with an example. [2]**

**(ii) Predict the output:-** **[2]**

```
class x
{void main()
{ int a= -2, b= -1;
double c= - -a-b%2/++a+2*a;
System.out.println(a);  }}
```

**(iii) Consider the following java statement and answer the questions that follow:** **[4]**

**String str="Washinton";    String str2= "Los Angeles";**

**(a) What is the result of System.out.println(str.substring(str2.indexOf('n',4)));**

**(b) What is the result of System.out.println(str.compareTo(str2));**

**(iv) A method is a program module that is used simultaneously at different instances in a program. It helps a user to reuse the same module multiple times in the program. Whenever you want to use a method in your program, you need to call it through its name. Some of the components/features of a method are as described below:** **[4]**

**(a) It defines the scope of usage of a method in the user's program.**

**(b) It is the value passed to the method at the time of its call.**

**(c) It is a return type that indicates that no value is returned from the method.**

**(d) It is an object oriented principle which defines method overloading.**

**Write the appropriate term used for each component/feature described above.**

**(v) The following program is checking if the entered number is an Evil number or Odious number. An evil number is a non-negative number that has an even number of 1's in its binary expansion. Numbers that are not Evil are called Odious Numbers. Examples :**

**Input : 3**

**Output : Evil Number**

**Explanation: Binary expansion of 3 is 11, the number of 1s in this is 2 i.e even.**

**Input : 16**

**Output : Odious Number(not an evil number)**

**Explanation: Binary expansion of 16 = 10000, having number of 1s =1 i.e odd.**

**Substitute the numbered places with suitable java statement(s)/expression so that the program can run.** **[4]**

```
import java.util.Scanner;
public class EvilNumber
{    public static void main( ) {
      Scanner in = new Scanner(System.in);
      System.out.print("Enter a positive number: ");
      int n = in.nextInt();
        if (n < 0)
       { System.out.println("Invalid Input");   return;     }
        int count = 0;    int p = 0;    int binNum = 0;
        while (n > 0)
       {    int d = ___?1?___;
            if (d == 1)
            ___?2?___;
            binNum += (int)(d * Math.pow(__?3?__, p));
            p++;        n /= 2;     }
        System.out.println("Binary Equivalent: " + binNum);
        System.out.println("No. of 1's: " + count);
        if (__?4?___== 0)           System.out.println("Output: Evil Number");
        else            System.out.println("Output: Not an Evil Number");    }}
```

**(vi) Predict output:** **[2]**

```
public class Pattern
{    public void display( )
   {   for (int i = 1; i < 10; i = i + 2)
      {   if(i==3) continue;
         for (int j = i; j > 0; j = j - 2)
         {    if(j==3) break;
             System.out.print(j + " ");        }
             System.out.println();       }    }  }
```

**(vii) What are the final values stored in variable x and z:** **[2]**

```
double a= - 2.26;       double b=5.74;       double y=26.4;
double x=Math.cbrt(Math.ceil(y));
double z=Math.max(Math.floor(x),y);
```

## Section B (60 Marks)

*(Answer any four questions from this Section.)*

***The answers in this section should consist of the programs in either BlueJ environment or any program environment with java as the base.***

***Each program should be written using variable description/mnemonic codes so that the logic of the program is clearly depicted. Flowcharts and algorithms are not required.***

**Question 3** **[15]**

**A goods rental company rents out Air conditioner on daily basis. The rental charges are mentioned below. Write a Java program with the following specifications:**

**Class :ENCool**

**Data members:**

| | |
|---|---|
| **c_name** | **: customer's name** |
| **add** | **: address of customer** |
| **days** | **: number of days hired** |

**Methods:**

| | |
|---|---|
| **ENCool()** | **: assigns data members to 0 or null** |
| **ENCool(………)** | **: a parameterized constructor to initialize data members** |
| **double generateBill()** | **: computes and returns bill based on the following slab-** |

| **Days** | **Rate** |
|---|---|
| **Upto 7 days** | **₹ 80/day** |
| **Next 7 days** | **₹ 70 /day** |
| **Next 7 days** | **₹ 60/day** |
| **Next 7 days** | **₹ 50/day** |
| **Above 28 days** | **₹ 45/day** |

**void display()** **: displays all the data members.**

**A discount of 15% on the bill amount is given if the Air conditioner is taken on rent for more than one month. Write a main method to create an object of the class and call the above member methods.**

**Question 4** **[15]**

**Write a program to input a sentence and remove the consecutive repeated characters of each word by replacing the sequence of repeated characters by its single occurrence. Also convert the first letter of the word to uppercase.**

**Example:**

**INPUT-Jaaavvvvvvvvaaaaaaaaaaa iiiis a funnnnyy languaaagee**

**OUTPUT – Java Is A Funy Language**

**INPUT–Heee iiiiss ggoiinggg**

**OUTPUT – He Is Going**

**Question 5** **[15]**

**Design a class to overload function display( ) as follows:**

i) **void display(String str, char ch) — checks whether the word *str* contains the letter *ch* at the beginning as well as at the end or not. If present, print 'Special Word' otherwise print 'No special word'.**

ii) **void display(String str1, String str2) — checks and prints whether both the words are equal or not.**

iii) **void display(String str, int n) — prints the character present at $n^{th}$ position in the word str.**

**Write a suitable main() method.**

**Question 6** **[8+7]**

**Design a menu-driven program to display the following as per the user's choice. Accepting the choice as 'A' for the first and 'B' for the second.**

**i) Write a program to display first five uppercase letters concatenated with last five lowercase letters.**

**ii) Write a program in Java to display the following patterns:**

**R**
**ER**
**TER**
**UTER**
**PUTER**
**MPUTER**
**OMPUTER**
**COMPUTER**

**Question 7** **[15]**

**Design two separate programs to do the following:**

**i) Write a program in java to check if two numbers taken as input are co-prime or not.**

**Two numbers are said to be co-prime, if their HCF is 1 (one)**

**Sample Input: 14, 15**

**Sample Output: They are co-prime.**

**ii) An abundant number is a number for which the sum of its proper divisors (excluding the number itself) is greater than the original number. Write a program to input number and check whether it is an abundant number or not.**

**Sample input: 12**

**Sample output: It is an abundant number.**

**Explanation: Its proper divisors are 1, 2, 3, 4 and 6**

**Sum = 1 + 2 + 3 + 4 + 6 = 16**

**Hence, 12 is an abundant number.**

**Question 8** **[15]**

**Write a program that encodes a word into Piglatin. To translate word into a piglatin word, convert the word into uppercase and then place the first vowel of the original word as the start of the new word along with the remaining alphabets. The alphabets present before the vowel being shifted towards the end followed by "AY".**

**Sample input: LONDON      Sample output: ONDONLAY**

**Sample input: Olympics      Sample output: OLYMPICSAY**